

deepening: the log of capital investment flow per worker and the log capital to labor ratio.

Using these data, we fit stacked first-difference industry task shift models of the form,

$$(14) \quad \Delta T_{jk\tau} = \alpha + \delta_{70-80} + \delta_{80-90} + \delta_{90-98} + \varphi CI_{j\tau} + \theta KI_{j\tau} + \varepsilon_{jk\tau},$$

where $CI_{j\tau}$ is industry j 's real log investment in computer capital per FTE over the contemporaneous decade in industry, $KI_{j\tau}$ is the analogous measure for real capital investment, the δ 's are time dummies equal to one in each of the decades post-1960 corresponding to their subscripts, and α is a common intercept. In this equation, the δ 's measure the trend change in industry task input in the 1970s, 1980s, and 1990s relative to the base period of the 1960s.

Estimates of equation (14) are found in Table IV. Two sets of Huber-White standard errors are tabulated for each model. Those in parentheses account for the fact that the NIPA capital measures are observed at a more aggregate level than the dependent variables measured from the CPS and Census (42 sectors versus 123 sectors for this exercise). The standard errors in brackets additionally account for potential serial correlation in industry task changes over succeeding decades (cf. Bertrand, Duflo, and Mullainathan [2004]).

As is visible in the table, the NIPA measure of computer investment consistently predicts relative declines in industry input of both routine cognitive and manual tasks and growth in input of nonroutine analytic and interactive tasks. How large are these relationships? We can gauge the model's explanatory power by comparing the magnitude of the estimated δ 's conditional on computer investment with the unconditional within-industry trends in task input observed for each decade. To facilitate this comparison, the bottom panel of Table IV tabulates the unconditional decadal trends. As with the Table III estimates, we find that industries making relatively greater investments in computer capital are responsible for the bulk of the observed substitution away from routine cognitive and manual tasks and toward nonroutine analytic and interactive tasks. Holding computer investment constant, we can explain more than 100 percent of the overall trend increase in nonroutine cognitive/analytic task input, a substantial part of the trend increase in nonroutine cognitive/interactive input, and substantial parts of the trend decreases in routine cognitive and routine manual inputs.

TABLE IV
COMPUTER INVESTMENT, CAPITAL INTENSITY, AND TASK INPUT IN THREE-DIGIT
INDUSTRIES 1960–1998: STACKED FIRST-DIFFERENCE ESTIMATES
DEPENDENT VARIABLE: $10 \times$ ANNUAL CHANGE IN QUANTILES OF TASK MEASURE,
MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

	A. Δ		B. Δ		C. Δ		D. Δ	
	Nonroutine analytic		Nonroutine interactive		Routine cognitive		Routine manual	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
$\text{Log}(CI/L)$	6.65 (4.13) [6.36]	6.76 (3.97) [5.90]	11.59 (3.21) [3.97]	10.03 (3.31) [4.50]	-8.27 (3.63) [4.74]	-8.30 (3.26) [3.76]	-9.11 (2.57) [3.27]	-8.20 (2.29) [2.86]
$\text{Log}(KI/L)$	1.22 (4.36) [6.36]		-3.41 (4.58) [4.45]		-2.93 (4.76) [7.31]		-2.42 (3.95) [5.87]	
$\Delta \text{Log}(K/L)$		0.24 (2.35) [2.36]		3.01 (2.24) [2.19]		-1.32 (2.12) [2.18]		-3.89 (1.92) [2.38]
1970–1980 dummy	-0.64 (1.08) [1.02]	-0.54 (1.29) [1.23]	1.38 (1.50) [2.07]	2.49 (1.62) [2.12]	-0.32 (1.31) [1.13]	-0.84 (1.58) [0.93]	0.68 (0.96) [0.94]	-0.80 (1.17) [1.03]
1980–1990 dummy	-0.34 (1.57) [1.43]	-0.25 (1.60) [1.67]	0.58 (1.81) [1.58]	1.83 (1.70) [1.46]	-1.62 (1.56) [1.33]	-2.14 (1.86) [1.35]	-1.32 (1.11) [0.86]	-2.90 (1.38) [1.07]
1990–1998 dummy	-1.19 (1.55) [1.77]	-1.13 (1.62) [1.93]	-1.91 (1.83) [1.85]	-0.90 (1.70) [1.88]	-1.33 (1.66) [1.93]	-1.71 (1.64) [1.63]	-1.15 (1.32) [0.95]	-2.36 (1.47) [1.13]
Intercept	8.89 (4.08) [5.45]	8.23 (4.42) [6.38]	12.40 (4.25) [4.76]	11.30 (3.59) [4.80]	-9.29 (4.12) [4.48]	-7.09 (3.63) [4.36]	-9.62 (3.14) [3.75]	-5.55 (2.67) [3.30]
R^2	0.06	0.06	0.11	0.12	0.14	0.14	0.20	0.21
Weighted mean of dependent variable								
1960–1970	1.16		1.74		1.30		1.63	
1970–1980	1.23		4.59		-0.20		0.98	
1980–1990	2.07		4.69		-2.05		-1.74	
1990–1998	2.15		3.76		-3.03		-2.82	

$n = 492$. Robust standard errors in parentheses are heteroskedasticity consistent and account for clustering of errors within 42 consistent NIPA sectors in each decade (168 clusters). Standard errors in brackets additionally account for potential serial correlation within sectors (42 clusters). Each column presents a separate OLS regression of ten times annual industry changes in task input on the indicated covariates. Sample is 123 consistent CIC industries, with four observations per industry. 1960–1970 and 1970–1980 changes use Census IPUMS samples, and 1980–1990 and 1990–1998 use CPS MORG samples. Estimates are weighted by mean industry share of total employment (in FTEs) over the endpoints of the years used to form the dependent variable. All capital measures are in millions of real 1996 dollars.

$\text{Log}(CI/L)$ and $\text{Log}(KI/L)$ are, respectively, one-tenth the log of annual computer investment per FTE and total capital investment per FTE between the two end years used to form the dependent variable. Means of $\text{Log}(CI/L)$ are -1.08, -0.95, -0.87, and -0.73 in 1960–1970, 1970–1980, 1980–1990, and 1990–1998, respectively. Means of $\text{Log}(KI/L)$ are -0.57, -0.54, -0.54, and -0.52 in the corresponding years.

$\Delta \text{Log}(K/L)$ is ten times the annual change in log capital/FTE over the two end years used to form the dependent variable. Means are 0.43, 0.10, 0.10, and 0.24 in the corresponding years.

A notable pattern in Table IV is that the coefficients on the capital investment and capital intensity variables are economically small and in most cases insignificant. This indicates that aggregate capital deepening—apart from computer investment—explains little of the observed change in task input.²² We have also explored a large number of variations of these models including estimating separate models for manufacturing and nonmanufacturing industries and for male and female task input; subdividing capital investment into equipment and structures investment; subdividing computer investment into hardware and software investment; and controlling for industries' log output, capital-output ratios, and import and export penetration. These tests confirm the overall robustness of our results.

V. TASK CHANGE WITHIN EDUCATION GROUPS AND OCCUPATIONS

The analyses above establish that industries undergoing rapid computerization reduced labor input of routine cognitive and manual tasks and increased labor input of nonroutine interactive and analytic tasks. Since better educated workers are likely to hold a comparative advantage in nonroutine versus routine tasks, one interpretation of these results is that they confirm the established pattern of increasing relative educational intensity in computerizing industries over the past several decades. While we do not disagree with this interpretation, our task framework makes a broader claim, namely that changes in the demand for workplace tasks, stemming from technological change, are an underlying cause—not merely a reflection—of relative demand shifts favoring educated labor. To test this broader implication, we exploit the unique features of the DOT to analyze two novel margins of task change: changes within education groups and changes within occupations.

V.A. Within-Industry Task Shifts by Education Group: 1980–1998

We showed in Tables III and IV that increased industry computerization predicts increased nonroutine cognitive activity and reduced routine cognitive and manual activities. Why does

22. This pattern echoes the findings of Berman, Bound, and Griliches [1994], Autor, Katz, and Krueger [1998], and Bresnahan, Brynjolfsson, and Hitt [2002] for skill upgrading.

this occur? One possibility is that as industries purchase computer capital, they hire better educated workers who specialize in these tasks. Alternatively, industries may change the task assignments of workers with given educational attainments, reducing their allocation to routine tasks and raising it to nonroutine tasks. We explore the relative importance of these two channels by estimating a variant of equation (13) for within-industry task upgrading by education group. Specifically, we estimate the model,

$$(15) \quad \Delta T_{ijk\tau} = \alpha_i + \phi_i \Delta C_j + \varepsilon_{ijk\tau},$$

where the dependent variable is the within-industry change in the mean of each DOT task, measured in centiles of the 1960 distribution, among workers of the same educational attainment. In this equation, i indexes each of four education groups—high school dropouts, high school graduates, some-college completers, and college graduates—and subscripts j , k , and τ refer to industries, tasks and time periods as above. We estimate this model using industry task data for 1980–1998 to exploit the (almost) contemporaneous industry computer use data for 1984–1997.

To establish a baseline for comparison, we initially estimate equation (15) for aggregate within-industry task changes over 1980–1998 (i.e., incorporating both between- and within-education group task shifts). Consistent with earlier findings, these estimates in panel A of Table V show striking correlations between industry computerization, rising labor input of routine cognitive and manual tasks, and declining labor input of nonroutine interactive and analytic tasks.

Panels B through E of Table V present analogous models estimated separately for the four education groups. Here, measured changes in task input stem solely from within-education group shifts in occupational distributions within industries. These estimates reveal that industry-level computerization is strongly predictive of shifts toward nonroutine and against routine tasks within essentially all education groups. For the two groups at the middle of the education distribution—high school graduates and those with some college—changing employment patterns within rapidly computerizing sectors entirely account for observed task shifts. More precisely, holding computer adoption fixed, our estimates would not predict any significant within-industry task change for either education group.

For the education groups at the bottom and top of the distri-